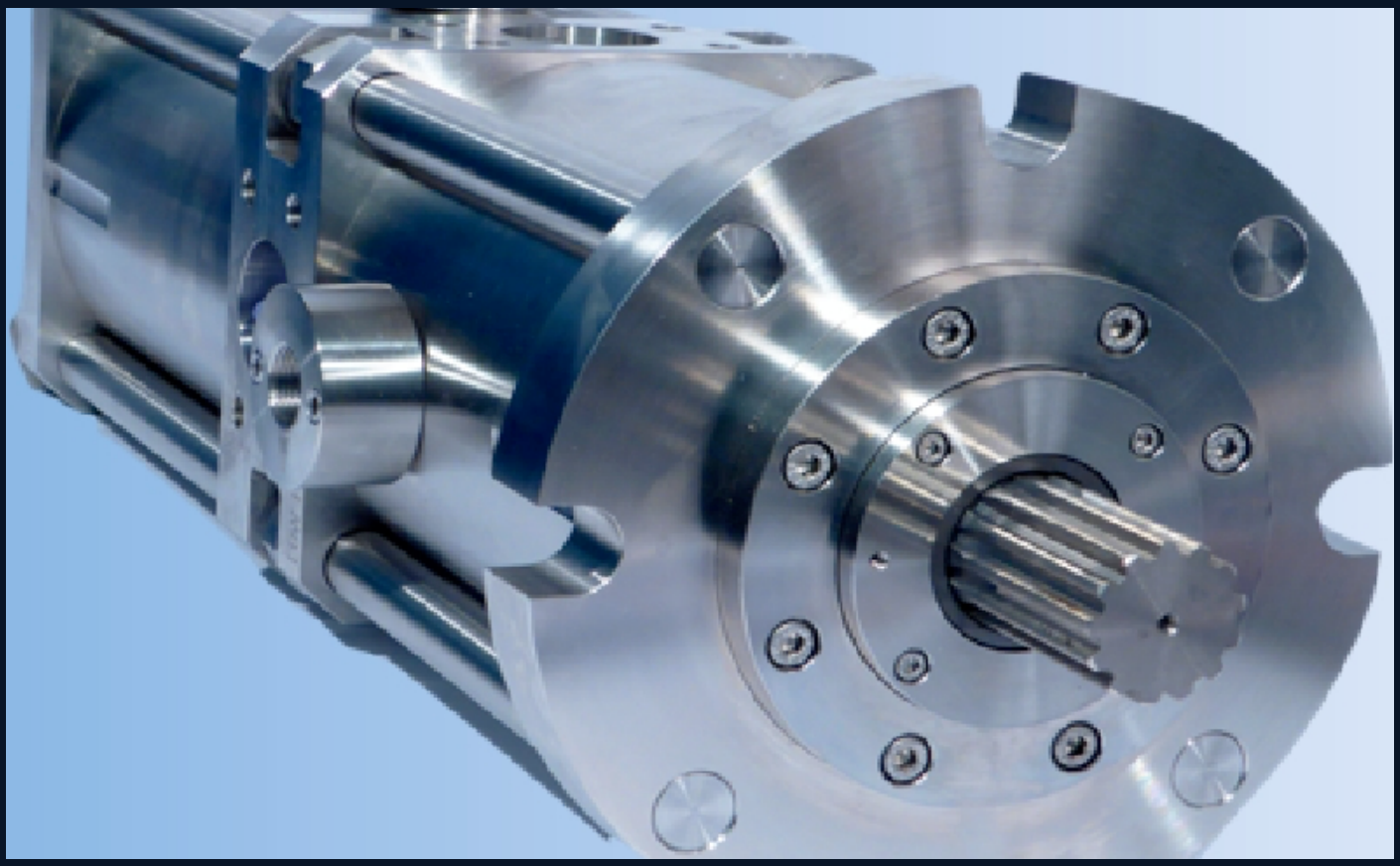
**WaterQuest Solutions**
STOP LEAKING OIL INTO THE FUTURE.

POWER CENTRE / REVERSE OSMOSIS

# Recover the pressure. Reduce the demand.

High-pressure feed pumping and brine energy recovery.
One integrated assembly.



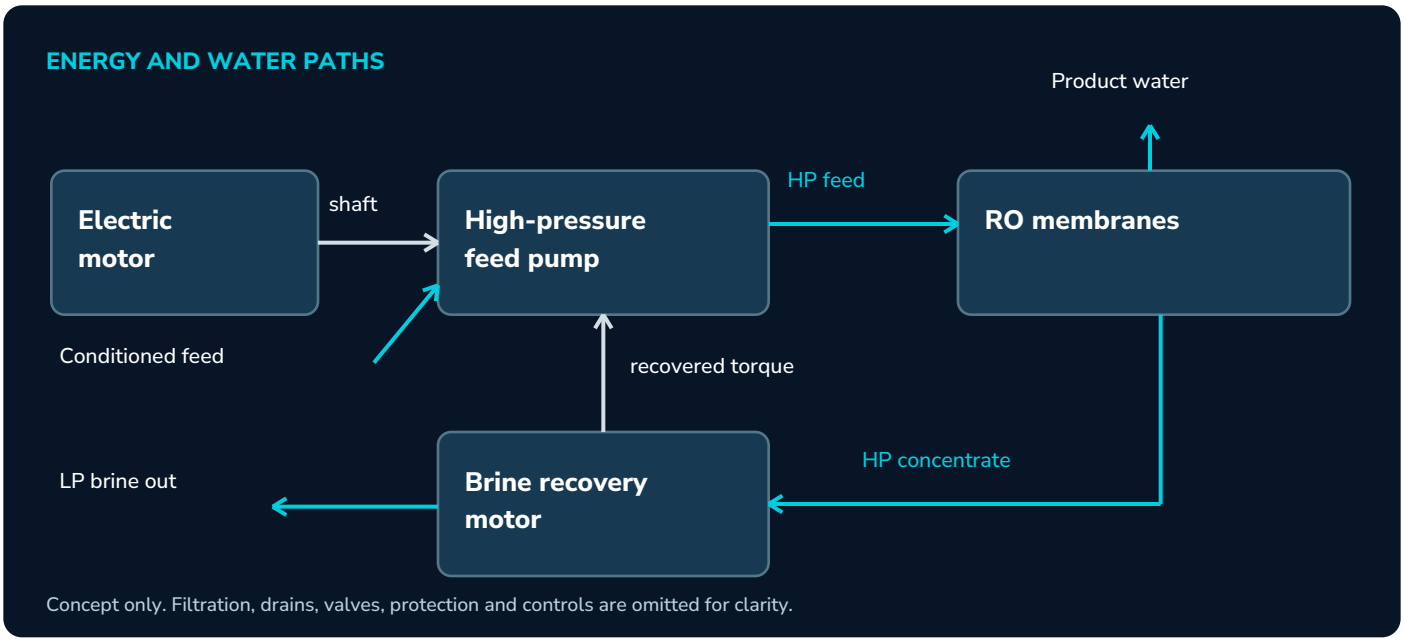
Power Centre pictured in WaterQuest product literature.

## Put the energy in your concentrate stream back to work.

RECOVER USEFUL ENERGY

# The brine still has work to do.

RO separates fresh water from a pressurised feed. The concentrate leaving the membranes still carries pressure energy. The Power Centre returns part of that energy to the pump drive.




POTENTIAL SAVINGS

## Up to 70%

versus systems without energy recovery

## One drive. Shared effort.

The electric motor supplies the remaining shaft input. Integrating the pump and recovery motor avoids a separate electric drive for energy recovery and a separate high-pressure recovery booster.

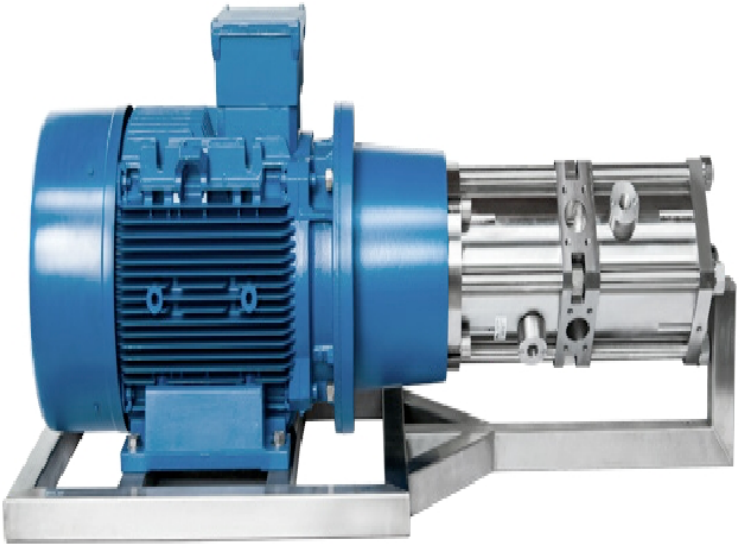
The manufacturer describes potential savings up to 70%. Actual savings depend on pressure, salinity, temperature, recovery and efficiency. Compare the same net water output and include feed pumping and other auxiliaries.

Basis: manufacturer PC300 product information. Energy claim is a comparison potential, not a guaranteed plant saving.

FEWER SEPARATE ASSEMBLIES

# Make room for a simpler installation.

A shared drive arrangement brings the high-pressure pump and energy-recovery motor together. Fewer separate machines can reduce the space, supports and interconnections needed for the recovery function.



A COORDINATED ASSEMBLY

01 **Electric drive motor**

02 **Power Centre pump + recovery motor**

03 **Bellhousing and flexible coupling**

04 **Mounting frame and support**

Assembly photograph from the supplied WaterQuest literature. Final layout varies by model.

## Compact by design

Useful where plant space matters: ships, offshore installations and packaged RO systems. Assess the complete installed footprint, including access and auxiliary equipment.

## Water-lubricated operation

The process water lubricates the pump and recovery-motor working surfaces. This removes hydraulic lubricating oil from those wetted units and the associated oil-change task.

Motor bearings and other plant equipment retain their own lubrication and maintenance requirements.

CONNECT THE COMPLETE CIRCUIT

# Four process connections. A complete plant around them.

Separate the integrated high-pressure recovery function from the upstream feed supply. The plant still needs the right inlet conditions, filtration, protection, drainage and controls.

| | |
|---|---|
| LOW-PRESSURE FEED IN | Conditioned feed enters the high-pressure pump. |
| HIGH-PRESSURE FEED OUT | Pressurised feed goes to the RO membrane train. |
| HIGH-PRESSURE BRINE IN | Membrane concentrate drives the recovery motor. |
| LOW-PRESSURE BRINE OUT | Depressurised concentrate leaves through the specified backpressure and discharge arrangement. |

These are the four main process paths. Case drains, pressure-switch ports and other auxiliary connections must also follow the selected installation drawing.

## Protect the inlet

For the published PC300 arrangement, maintain at least 3.5 bar (50.8 psi) at the inlet pressure-switch port. Low inlet pressure initiates shutdown. Select filtration and feed/boost capacity for the actual flow.

## Mount and connect correctly

Use the specified horizontal mounting, flexible coupling, rear support, materials and port arrangement. Route case drains as instructed. Maintain pressure protection and instrumentation for commissioning and operation.

Speed, pressure, temperature and connection limits are model-specific. Use the current approved drawing and quotation.

MATCH THE DUTY. PLAN THE SERVICE.

# Configure around the water you need.

Specify the feedwater and the required product-water output together. Recovery is selected during configuration; motor speed then controls flow within the approved operating envelope.



Power Centre range photograph from the supplied WaterQuest literature.

## Water and materials

Review salinity, temperature, fouling potential and cleaning chemicals. Duplex stainless options and seals must suit the feed and concentrate. Confirm the material schedule for the selected unit.

## Accessible servicing

Allow space for inspection and removal. Water lubrication avoids pump oil changes, but filtration, flushing, seals and mechanical inspection remain part of the maintenance plan.

**Service intervals:** supplied range literature describes up to 8,000 hours when its specifications are met. Confirm the applicable interval, water quality and service or exchange-unit availability for the quoted configuration.

YOUR NEXT DESALINATION PROJECT

# Select the unit. Define the result.

Explore the current model pages for performance information and installation details. A model name is not a guaranteed daily water-production rating.

## PC250

Power Centre / View model online

## PC300

Power Centre / View model online

## PC500

Power Centre / View model online

## PC1000

Power Centre Plus / View model online

The supplied range overview describes eight sizes and production spanning 3-1,000 m³/day (793-264,172 US gal/day). Treat that as a range enquiry guide; confirm current availability, feed conditions, operating hours and net output for the selected package.

BRING US THE DUTY

Feed analysis and seasonal temperature. Daily water demand and operating hours. Membrane pressure and recovery. Available power, installation space and service requirements.

WaterQuest provides Canadian selection and integration support to turn those inputs into an equipment scope and quotation.

**Discuss your Power Centre application**

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